

Personal Project Dataset Validation
Report
Minor AI for Society

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1. Introduction

Public datasets enable rapid experimentation with machine learning, but their suitability for real-world and constrained applications (e.g., TinyML) varies significantly. For this Minor AI for Society project, dataset validation is therefore a critical step to ensure reliable and deployable model performance.

This report evaluates four plant-related datasets to determine their suitability for lightweight machine learning models. Each dataset is assessed using a consistent workflow that combines data quality checks with practical model testing, including baseline models, compact neural networks (TinyMLP), regression variants, and anomaly detection.

The goal is to identify which dataset best supports accurate, efficient, and realistic deployment within constrained environments.

2. Purpose

This document summarizes the validation work for the four collected datasets and evaluates whether each dataset is suitable for the personal project goals.

The validation focuses on:

- feature availability for the target use case,
- basic data quality checks,
- baseline model behavior,
- compatibility with small models (TinyMLP / TinyRegression), and
- anomaly-detection potential.

3. Used Datasets

The validation in this report uses the following public datasets:

1. Plant Health Diseases Dataset (Kaggle): <https://www.kaggle.com/datasets/muqaddasejaz/plant-health-diseases-dataset>
2. Indoor Plant Health and Growth Dataset (Kaggle): <https://www.kaggle.com/datasets/souvikrana17/indoor-plant-health-and-growth-dataset>
3. Edge Assisted Agricultural Sensor Dataset (Kaggle): <https://www.kaggle.com/datasets/colabsss/edge-assisted-agricultural-sensor-dataset>
4. Plant Health Biosensor Dataset (Kaggle): <https://www.kaggle.com/datasets/programmer3/plant-health-biosensor-dataset>

4. Validation Workflow

The following steps were used for each dataset to keep the validation process consistent and reproducible:

1. Imported the raw dataset and inspected schema/column names.
2. Checked required fields for the project use case (sensor-like inputs + plant health target).
3. Reviewed data quality (missing values, duplicates, data types, and target distribution).
4. Performed basic feature preparation and train/test splitting.
5. Trained a baseline `RandomForestClassifier` to estimate practical signal strength.
6. Trained a compact `TinyMLP` model to evaluate TinyML feasibility.
7. Where applicable, trained a compact regression variant (`TinyRegression`) and recorded `MSE`, `MAE`, and `R2`.
8. Ran anomaly detection (`IsolationForest`) to evaluate anomaly-detection suitability.
9. Compared the outcomes across datasets to rank suitability for deployment constraints.

This workflow was applied to provide both data-quality validation and model-readiness validation before selecting a final dataset.

5. Validation Criteria

Each dataset was reviewed using the same checklist:

- Required fields available (light, soil moisture, humidity, ambient temperature, plant health or equivalent target).
- Missing values and duplicates.
- Data types and label format.
- Class balance / target signal quality.
- Baseline Random Forest test performance.
- TinyMLP / TinyRegression feasibility.

6. Dataset 1: Plant Health Data

6.1. Findings

- The dataset is one of the most usable options for the project.
- Raw features show imbalance in predictive power (single-feature dominance).
- After feature engineering, model behavior improves significantly and becomes more suitable for compact models.

6.2. Key Metrics

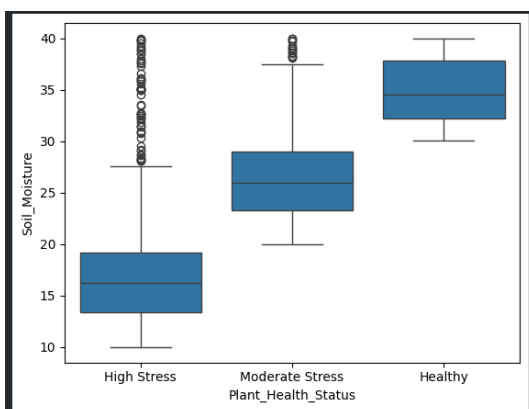
- Random Forest test accuracy: 0.8240
- TinyMLP test accuracy: 0.6958
- TinyRegression: MSE 0.2840, MAE 0.4148, R^2 0.5388

6.3. Engineered Version

The engineered version focuses on finding the relation between different raw values. Rather taking the raw data as is, it looks for how can it be used to improve the models feature importance.

6.4. Proof Images

```
Dataset Validation Summary
-----
Rows used: 1200
Features used: ['Light_Intensity', 'Soil_Moisture', 'Humidity', 'Ambient_Temperature']
Target classes: ['Healthy', 'High Stress', 'Moderate Stress']
Missing values removed: 0
Duplicate rows in original data: 0
Test accuracy: 0.8458
```



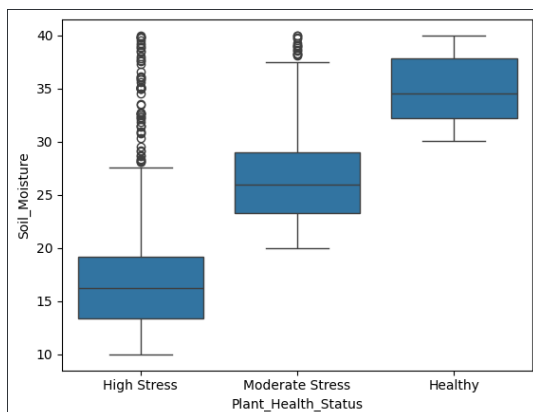
Train set report:				
	precision	recall	f1-score	support
Healthy	1.00	1.00	1.00	239
High Stress	1.00	1.00	1.00	400
Moderate Stress	1.00	1.00	1.00	321
accuracy			1.00	960
macro avg	1.00	1.00	1.00	960
weighted avg	1.00	1.00	1.00	960
Test set report:				
	precision	recall	f1-score	support
Healthy	0.73	0.98	0.84	60
High Stress	0.99	0.73	0.84	100
Moderate Stress	0.84	0.89	0.86	80
accuracy			0.85	240
macro avg	0.85	0.87	0.85	240
weighted avg	0.87	0.85	0.85	240

6.4.1. Engineered Version

```

Dataset Validation Summary
-----
Original rows: 1200
Rows used after cleaning: 1200
Features used: ['Soil_Moisture', 'humidity_stress', 'moisture_temp_interaction', 'light_moisture_interaction']
Target classes: ['Healthy', 'High Stress', 'Moderate Stress']
Missing values removed: 0
Duplicate rows in original data: 0
Train accuracy: 1.0000
Test accuracy: 0.8417

```



	precision	recall	f1-score	support
Healthy	1.00	1.00	1.00	239
High Stress	1.00	1.00	1.00	400
Moderate Stress	1.00	1.00	1.00	321
accuracy			1.00	960
macro avg	1.00	1.00	1.00	960
weighted avg	1.00	1.00	1.00	960
Test set report:				
	precision	recall	f1-score	support
Healthy	0.73	0.98	0.84	60
High Stress	0.99	0.73	0.84	100
Moderate Stress	0.82	0.88	0.85	80
...				
accuracy			0.84	240
macro avg	0.85	0.86	0.84	240
weighted avg	0.87	0.84	0.84	240

7. Dataset 2: Indoor Plant Health and Growth Factors

7.1. Findings

- Baseline performance is low and unstable.
- Adding `Plant_ID` / plant-specific information produces only marginal improvement.
- Correlation analysis indicates weak relationships between input features and the health target.
- Feature importances are relatively flat, suggesting low signal strength.
- Adding extra columns (for example light exposure, soil type, pest presence) did not improve results and in some runs reduced performance.

7.2. Key Metrics

- Random Forest test accuracy: `0.2140`
- TinyMLP test accuracy: `0.2040`
- Anomaly detections: `132`

7.3. Proof Images

Indoor Plant Health Summary

```
Dataset Validation Summary
-----
Rows used: 1000
Features used: ['Soil_Moisture', 'Humidity', 'Ambient_Temperature']
Target classes: ['Healthy', 'High Stress', 'Moderate Stress']
Missing values removed: 0
Duplicate rows in original data: 0
Test accuracy: 0.3600
```

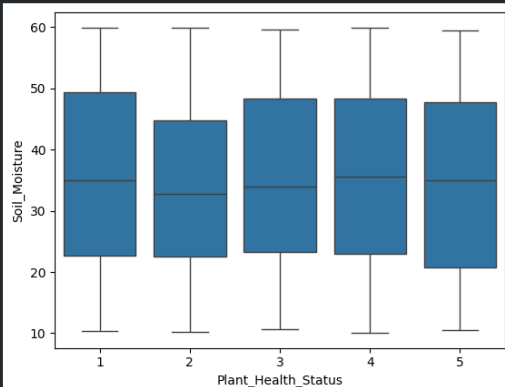
Indoor Plant Health With ID Summary

```
Dataset Validation Summary
-----
Rows used: 1000
Features used: ['Soil_Moisture', 'Humidity', 'Ambient_Temperature', 'Plant_ID']
Target classes: ['Healthy', 'High Stress', 'Moderate Stress']
Missing values removed: 0
Duplicate rows in original data: 0
Test accuracy: 0.3950
```

```

Soil_Moisture      0.003656
Humidity           0.031807
Ambient_Temperature 0.064863
Plant_Health_Status 1.000000
Name: Plant_Health_Status, dtype: float64

```



Train set report:

	precision	recall	f1-score	support
Healthy	1.00	1.00	1.00	170
High Stress	1.00	1.00	1.00	328
Moderate Stress	1.00	1.00	1.00	302
accuracy			1.00	800
macro avg	1.00	1.00	1.00	800
weighted avg	1.00	1.00	1.00	800

Test set report:

	precision	recall	f1-score	support
Healthy	0.40	0.14	0.21	43
High Stress	0.35	0.43	0.38	82
Moderate Stress	0.36	0.41	0.39	75
accuracy			0.36	200
macro avg	0.37	0.33	0.33	200
weighted avg	0.37	0.36	0.35	200

8. Dataset 3: Agriculture Dataset with Target

8.1. Findings

- The dataset can be used for broad experimentation, but predictive strength is limited.
- Baseline classification quality is low, indicating noisy or weakly informative feature-target mapping.
- TinyRegression results also indicate weak explanatory power ($R^2 < 0$).

8.2. Key Metrics

- Random Forest test accuracy: 0.3125
- TinyMLP test accuracy: 0.3640
- TinyRegression: MSE 0.7001, MAE 0.7157, R^2 -0.0730
- Anomaly detections: 218

8.3. Proof Images

Agriculture Dataset Summary

Dataset Validation Summary

Rows used: 2000

Features used: ['Soil_Moisture', 'Humidity', 'Ambient_Temperature']

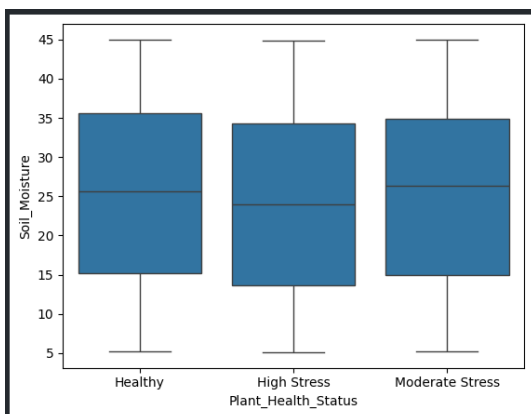
Target classes: ['Healthy', 'High Stress', 'Moderate Stress']

Missing values removed: 0

Duplicate rows in original data: 0

Test accuracy: 0.3125

Agriculture Dataset Correlation



Agriculture Dataset Report

Train set report:				
	precision	recall	f1-score	support
Healthy	1.00	1.00	1.00	543
High Stress	1.00	1.00	1.00	550
Moderate Stress	1.00	1.00	1.00	507
accuracy			1.00	1600
macro avg	1.00	1.00	1.00	1600
weighted avg	1.00	1.00	1.00	1600
Test set report:				
	precision	recall	f1-score	support
Healthy	0.36	0.40	0.38	136
High Stress	0.38	0.38	0.38	138
Moderate Stress	0.33	0.28	0.30	126
accuracy			0.36	400
macro avg	0.36	0.36	0.35	400
weighted avg	0.36	0.36	0.36	400

9. Dataset 4: Plant Health Biosensor Dataset

9.1. Findings

- The dataset shows a moderate predictive signal.
- It appears more aligned with sensor-driven use cases and may transfer better to constrained (TinyML-style) deployment than the weaker datasets.
- Requires the same quality checks as the others, but initial behavior is promising.

9.2. Key Metrics

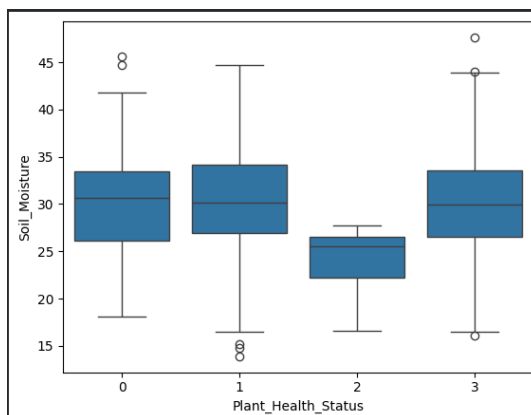
- Baseline test accuracy (notebook run): **0.6295**

9.3. Proof Images

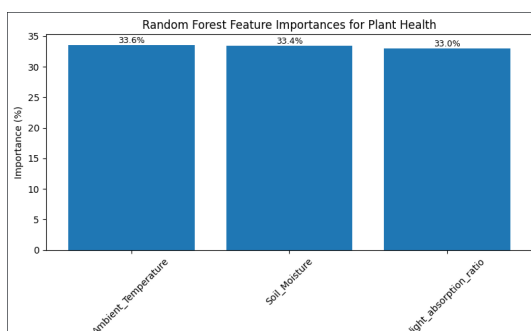
Plant health BioSensor Summary

```
Dataset Validation Summary
-----
Rows used: 1253
Features used: ['Soil_Moisture', 'light_absorption_ratio', 'Ambient_Temperature']
Target classes: [0, 1, 2, 3]
Missing values removed: 0
Duplicate rows in original data: 0
Test accuracy: 0.6295
```

Plant health BioSensor Correlation



Plant health BioSensor Feature Importance



10. Cross-Dataset Comparison (TinyML / TinyRegression / Anomaly Detection)

The combined comparison supports the final dataset ranking by showing how each dataset behaves under:

- baseline classification (**RandomForest**)
- compact neural classification (**TinyMLP**)
- compact regression (**TinyRegression**)
- anomaly detection (**IsolationForest**)

Datasets Comparison

	model	accuracy	mse	mae	r2	anomaly_count	dataset
0	RandomForest	0.825000	NaN	NaN	NaN	NaN	Plant_Health
1	TinyMLP	0.695833	NaN	NaN	NaN	NaN	Plant_Health
2	TinyRegression	NaN	0.285050	0.414802	0.538795	NaN	Plant_Health
3	AnomalyDetection	NaN	NaN	NaN	NaN	701.0	Plant_Health
0	RandomForest	0.215000	NaN	NaN	NaN	NaN	Indoor_Plant
1	TinyMLP	0.205000	NaN	NaN	NaN	NaN	Indoor_Plant
2	AnomalyDetection	NaN	NaN	NaN	NaN	132.0	Indoor_Plant
0	RandomForest	0.312500	NaN	NaN	NaN	NaN	Agriculture
1	TinyMLP	0.365000	NaN	NaN	NaN	NaN	Agriculture
2	TinyRegression	NaN	0.700069	0.715652	-0.072995	NaN	Agriculture
3	AnomalyDetection	NaN	NaN	NaN	NaN	218.0	Agriculture

The comparison image is used as supporting evidence that Plant Health Data (especially engineered) is the strongest candidate for lightweight deployment scenarios.

11. Executive Summary

Table 1. Executive Summary

Dataset	Baseline Signal	TinyML Readiness	Overall Verdict
Plant Health Data	Strong	Good	Recommended after feature engineering.
Indoor Plant Health & Growth Factors	Weak	Poor	Not recommended in current form (weak relationship to target).
Agriculture Dataset with Target	Weak-to-moderate	Limited	Usable for experiments, not strong for reliable prediction.
Plant Health Biosensor Dataset	Moderate	Promising	Potentially usable for constrained embedded models.

12. Final Recommendation

For the project objective (small, practical, and deployable predictive models), prioritize:

- 1. **Plant Health Data (engineered version)** as the primary dataset.
- 2. **Plant Health Biosensor Dataset** as a secondary candidate.

Use Agriculture as an exploratory dataset only, and avoid relying on Indoor Plant Health and Growth Factors for final model training unless substantial restructuring/label redesign is performed.